

PATHOLOGY OF UPPER AIRWAY

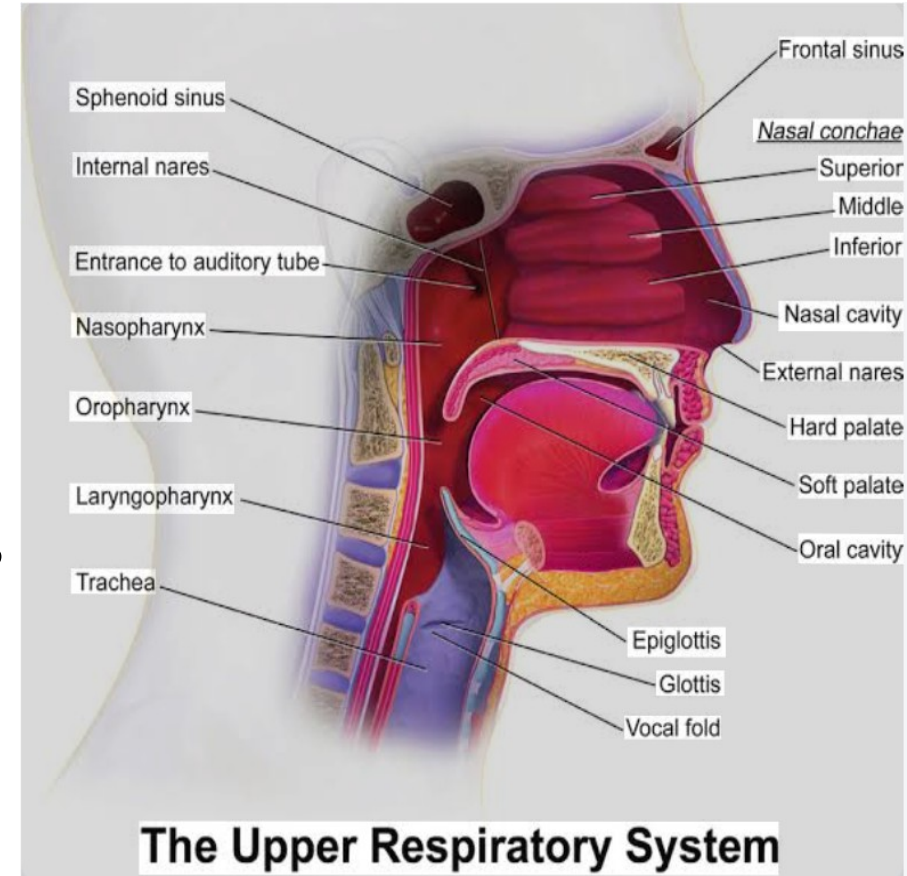
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OUTLINE

- Introduction
- Nose
- Pharynx
- Larynx
- Ear

Introduction

- The term upper airways include the nose, pharynx, and larynx and their related parts.
- Disorders of these structures are among the most common afflictions.
- The overwhelming majority of these disorders are more nuisance than threat.



NASAL CAVITY AND PARANASAL SINUSES

Nonneoplastic Diseases

Infectious Rhinitis

- ❖ Commonly referred to as “common cold”
- ❖ In most instances caused by one or more viruses.
 - Major offenders are adenoviruses, echoviruses, and **rhinoviruses**.
- ❖ They evoke a profuse **catarrhal discharge**.
- ❖ During the initial acute stages,
 - The nasal mucosa is thickened, edematous, and red
 - The nasal cavities are narrowed and the turbinates are enlarged
- ❖ These changes may extend, to produce **pharyngotonsillitis**
- ❖ Secondary bacterial infection enhances the inflammatory reaction and produces a **mucopurulent** or sometimes frankly **suppurative exudate**.



Allergic Rhinitis(hay fever)

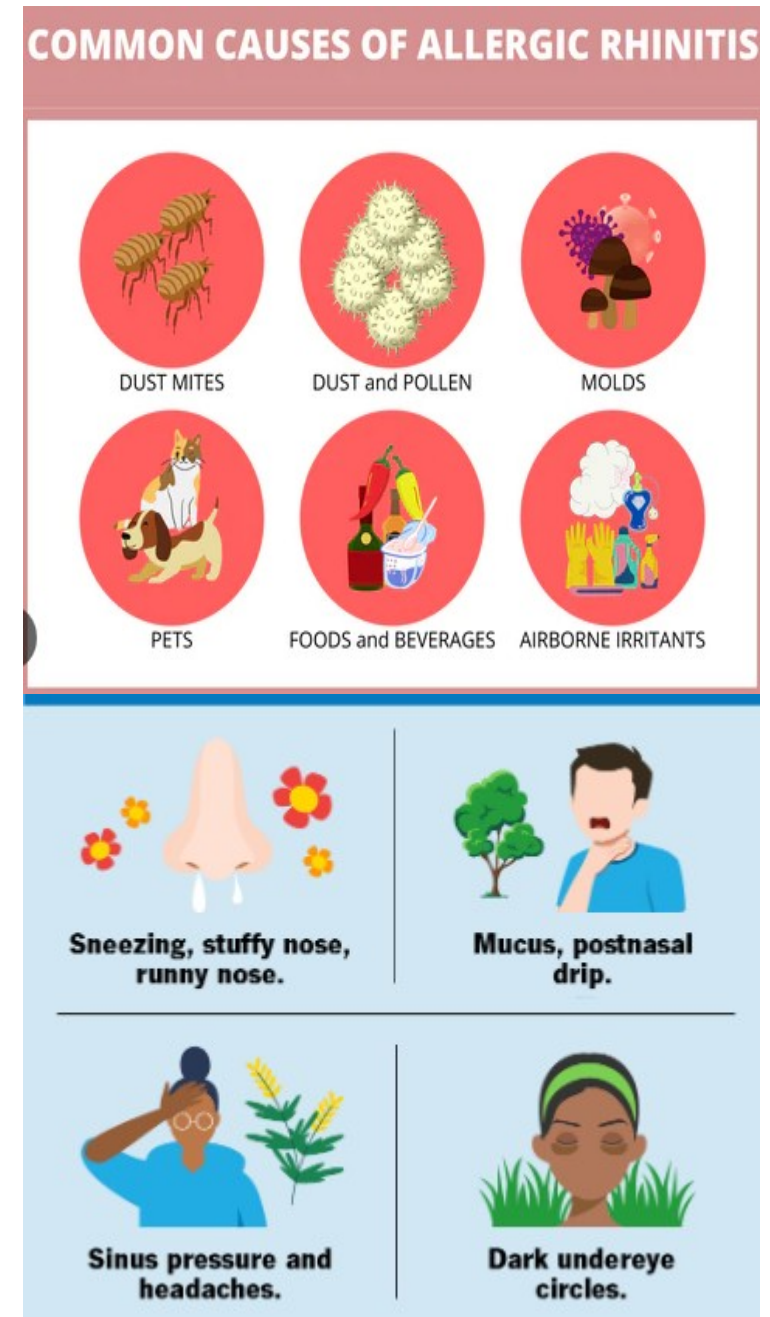
- ❖ Is initiated by hypersensitivity reactions to one of allergens
 - plant pollens, fungi, animal allergens, and dust mites.
- ❖ Is an IgE-mediated immune reaction with an early- and late phase response characterized by
 - marked mucosal edema, redness, and mucus secretion
 - leukocyte infiltration

Clinical Features

- **Cardinal Symptoms:** Paroxysmal sneezing, profuse watery rhinorrhea, nasal congestion, and nasal/pruritus of the nose, palate, and eyes.

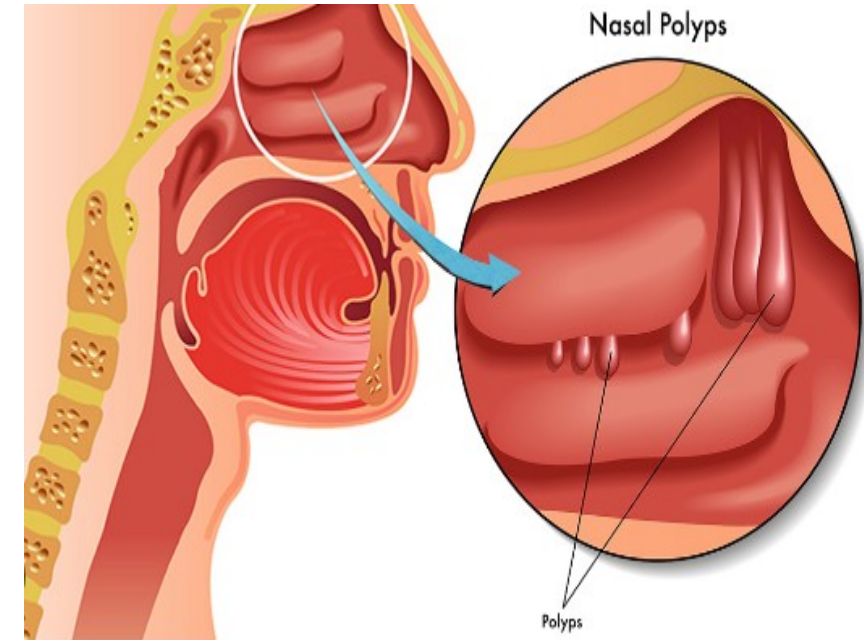
Patterns

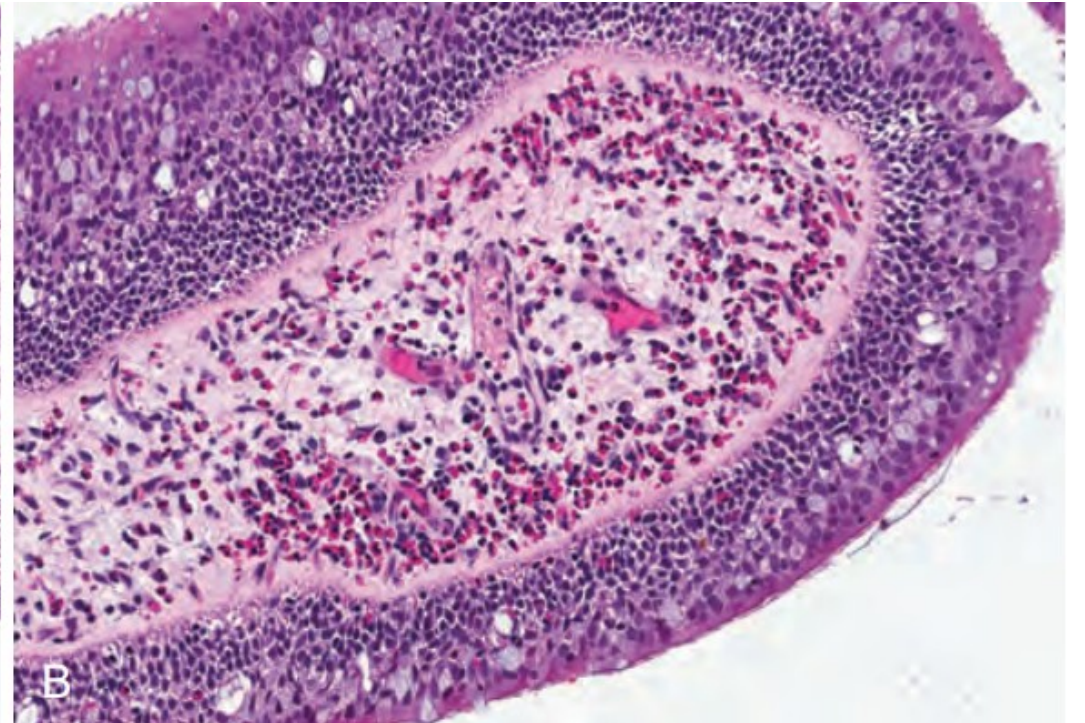
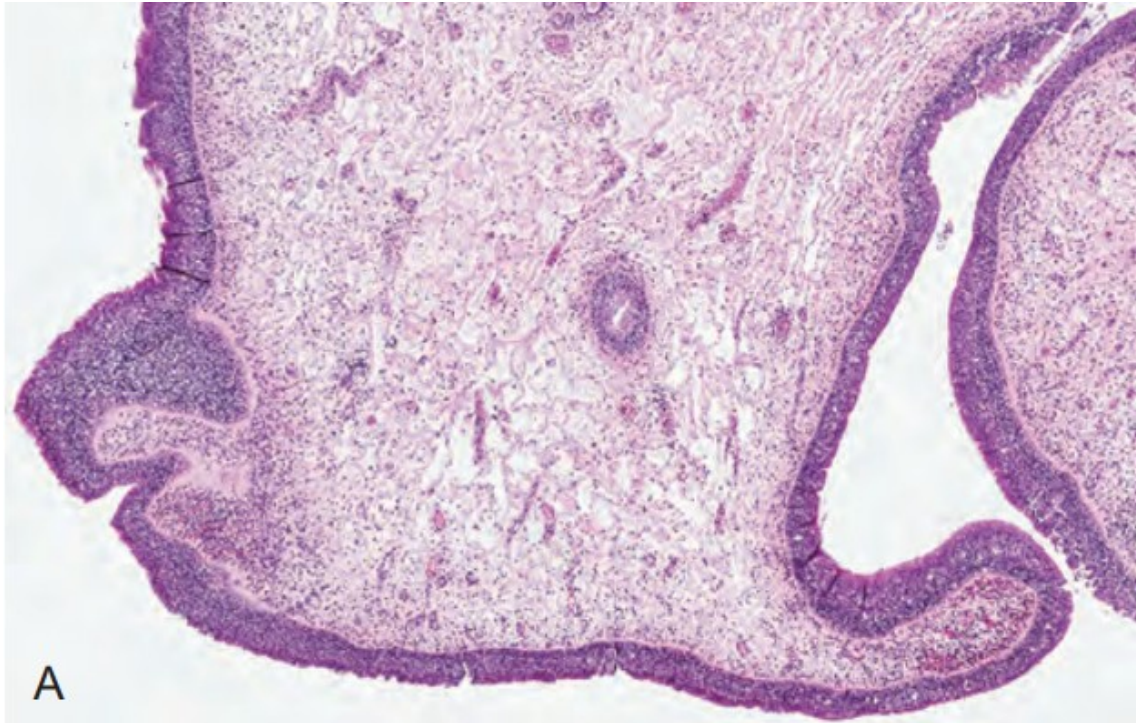
- **Seasonal:** Symptoms occur at **specific times of the year** (e.g., spring or fall) corresponding to pollen exposure.
- **Perennial:** Symptoms occur **year-round** due to constant exposure to indoor allergens like dust mites or pet dander.



Nasal Polyps

- ❖ Focal protrusions of the mucosa due to recurrent attacks of rhinitis
- ❖ Most arise from the **lateral nasal wall** or ethmoid recess
- ❖ May reach **3 to 4 cm** in length.
- ❖ On histologic examination
 - edematous mucosa having a loose stroma
 - hyperplastic or cystic mucous glands
 - infiltrated with a **variety** of inflammatory cells
- ❖ In the absence of bacterial infection, their mucosal covering is intact
- ❖ When multiple or large, the polyps may encroach on the airway and impair sinus drainage.
- ❖ Most people with nasal polyps are **not atopic**.
- ❖ Only **0.5% of atopic** patients develop polyps.



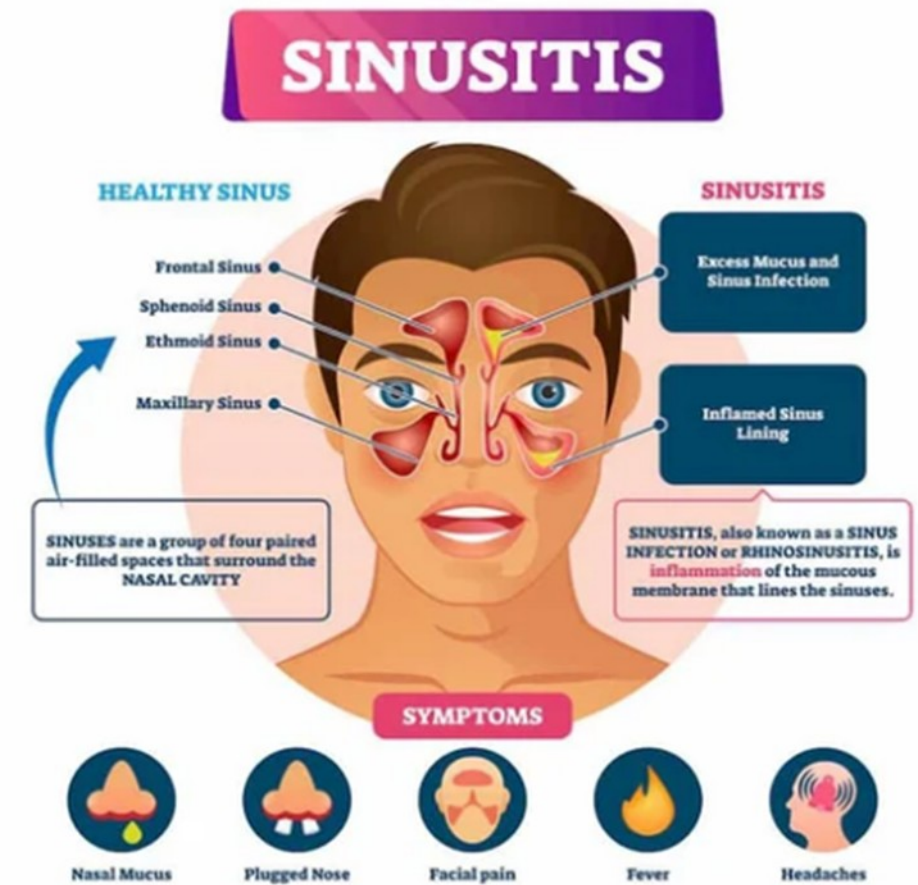


Chronic Rhinitis

- ❖ Is a sequel to repeated attacks of acute rhinitis (either microbial or allergic)
 - with the eventual development of superimposed bacterial infection.
- ❖ A **deviated nasal septum** or **nasal polyps** with impaired drainage of secretions contribute to the likelihood of microbial invasion.
- ❖ Frequently, there is
 - ❖ superficial desquamation or ulceration of the mucosal epithelium
 - ❖ a variable inflammatory infiltrate of neutrophils, lymphocytes, and plasma cells subjacent to the epithelium.
- These suppurative infections sometimes extend into the air sinuses.

Sinusitis

- ❖ Refers to inflammation of the mucous membranes of the paranasal sinuses.
- Acute sinusitis is a disorder of **less than 3 weeks'** duration,
 - caused predominantly by extension of infection from the nasal mucosa.
 - Haemophilus influenzae and ranhamella catarrhalis are the most common causes
 - **Maxillary** sinusitis may also be caused by **odontogenic** infections
- **Symptoms:** Purulent nasal discharge, nasal congestion, **facial pain or pressure** (worse when bending forward), headache, and fever.



❖ **Chronic sinusitis** is a sequel of acute inflammation, either as a result of

- incomplete resolution of infection or
- recurrent acute complications

❖ The purulent exudate in chronic sinusitis almost always includes **anaerobic** bacteria

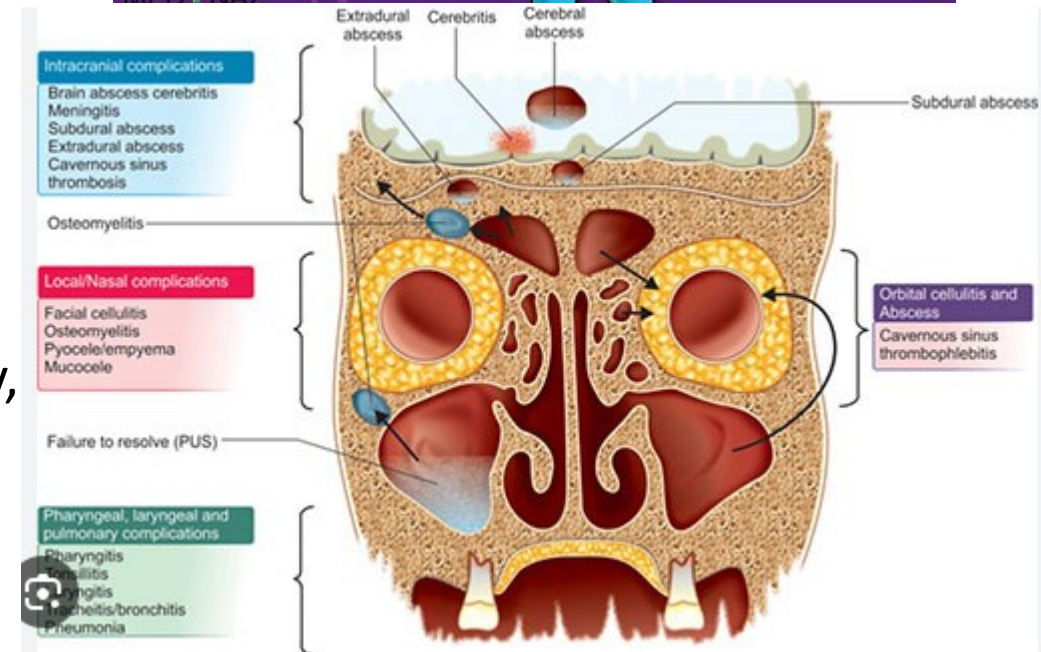
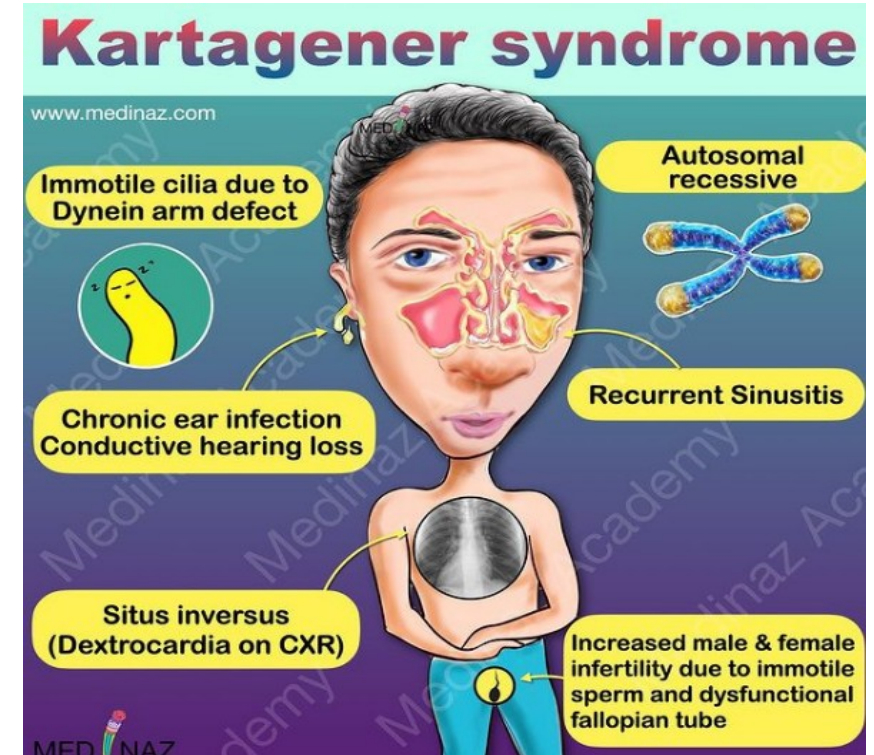
❖ Severe forms of chronic sinusitis are caused by fungi (e.g., mucormycosis)

Kartagener syndrome

- Sinusitis, bronchiectasis and situs inversus

❖ Acute or chronic sinusitis may be followed by a number of complications

- mucocoele (the accumulation of mucus)
- pyocoele (accumulation of mucopurulent exudate)
- osteomyelitis, septic thrombophlebitis and, ultimately, intracranial infections.



Necrotizing Lesions of the Nose and Upper Airways

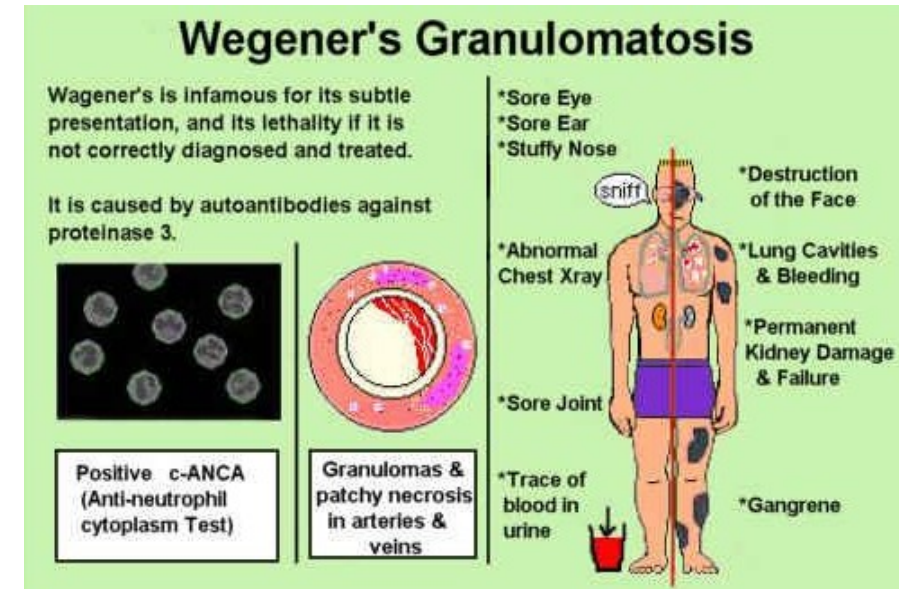
❖ Acute fungal infections (including mucormycosis)

- particularly in patients with diabetes and immunosuppressed patients

❖ Granulomatosis with polyangiitis(Wegener granulomatosis)

❖ Extranodal NK/T-cell lymphoma, nasal type,

- ❖ is a lymphoma in which the tumor cells harbor **EBV**
- ❖ typically seen in **males** who are in the **fifth or sixth** decade of life
- ❖ most commonly occur in individuals of Asian and Latin American descent.
- ❖ Ulceration and superimposed bacterial infections complicate the process.

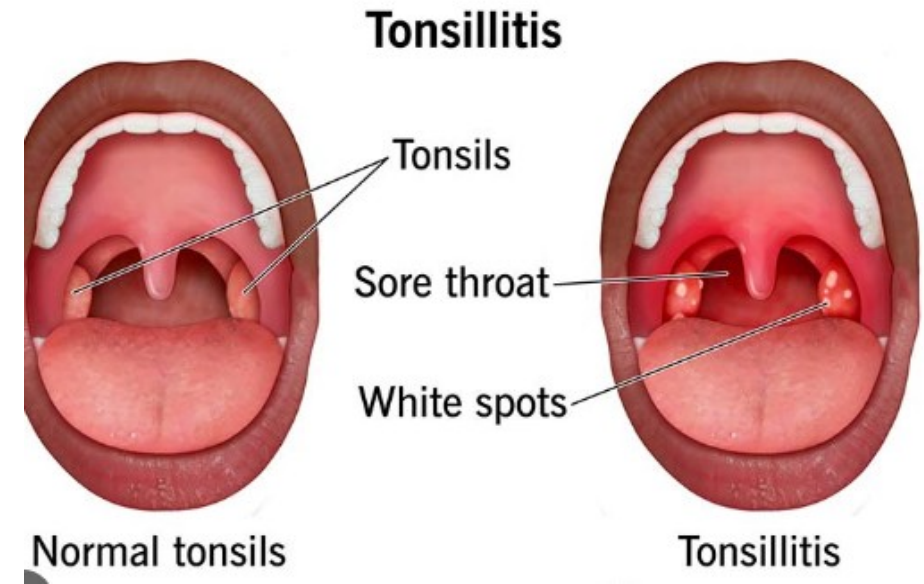


NASOPHARYNX

- ❖ The nasopharyngeal mucosa, related lymphoid structures, and glands may be involved in
 - Infections (e.g., diphtheria, infectious mononucleosis)
 - Neoplasms

Inflammatory Lesions

- ❖ Pharyngitis and tonsillitis frequently accompany viral upper respiratory infections.
 - Rhinoviruses, echoviruses, and adenoviruses are the most common causes
 - influenza or respiratory syncytial virus.
- ❖ Erythema and edema of the nasopharyngeal mucosa with reactive enlargement of nearby tonsils and lymph nodes are characteristic.



- ❖ Bacterial infection may cause pharyngitis and tonsillitis or, alternatively, occur as a secondary process superimposed on viral infection.
 - ❖ **β -hemolytic streptococci** are the most frequent pathogens
 - ❖ *Staphylococcus aureus* or other bacteria may also be present
- ❖ When bacteria are present,
 - ❖ exudate and exudative membranes (**pseudomembranes**) may cover the nasopharyngeal mucosa as well as enlarged nasopalatine and palatine tonsils.
- ❖ The major importance of streptococcal “sore throat” lies in the possible development of **late sequelae**, such as
 - ❖ rheumatic fever
 - ❖ glomerulonephritis

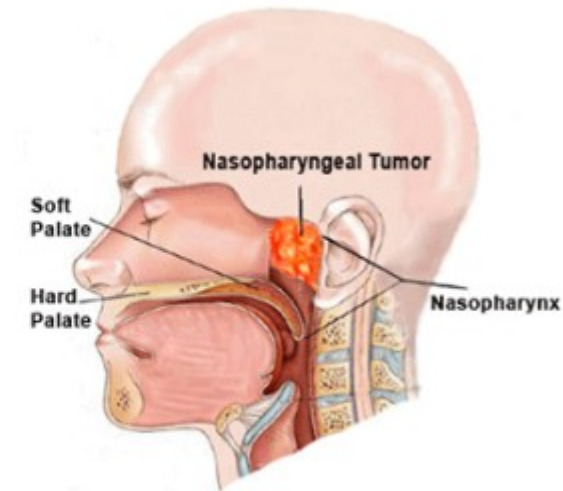


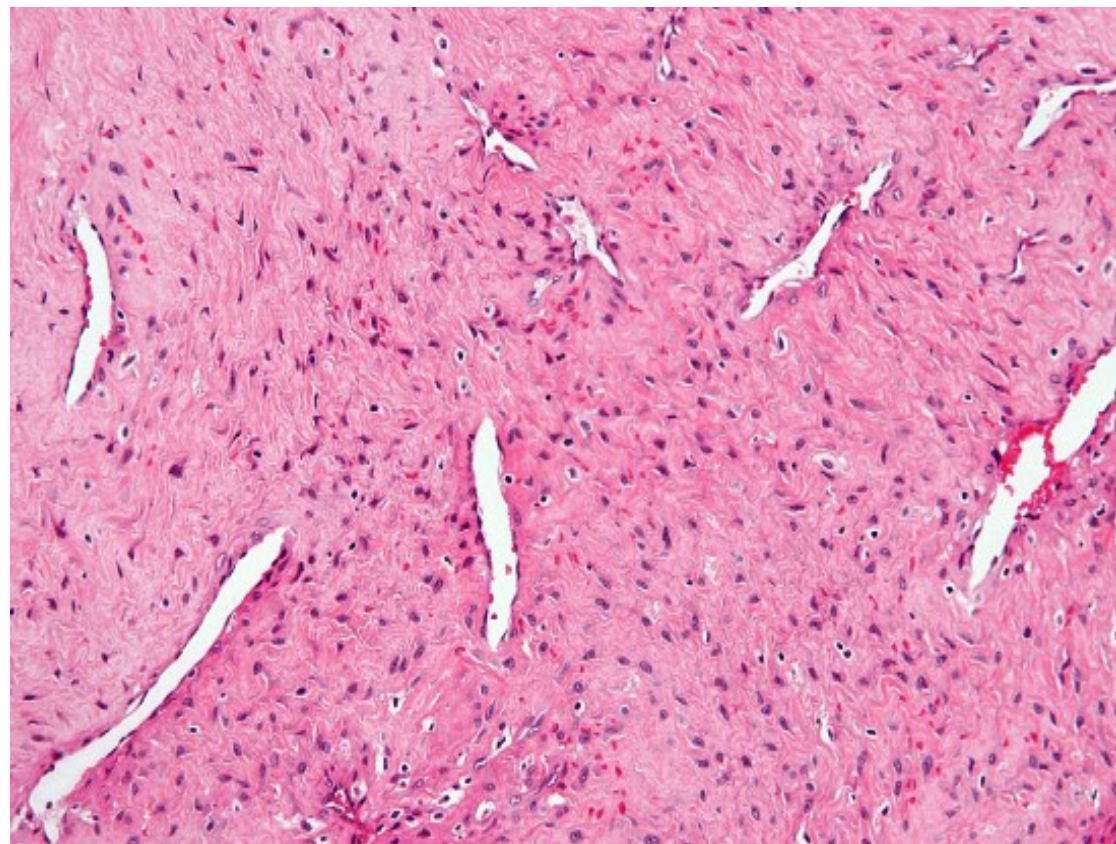
TUMORS OF THE NOSE, SINUSES, AND NASOPHARYNX

- ❖ Tumors in the nose, sinuses, and nasopharynx are **infrequent**.
- ❖ Include the wide spectrum of mesenchymal and epithelial neoplasms

Nasopharyngeal Angiofibroma

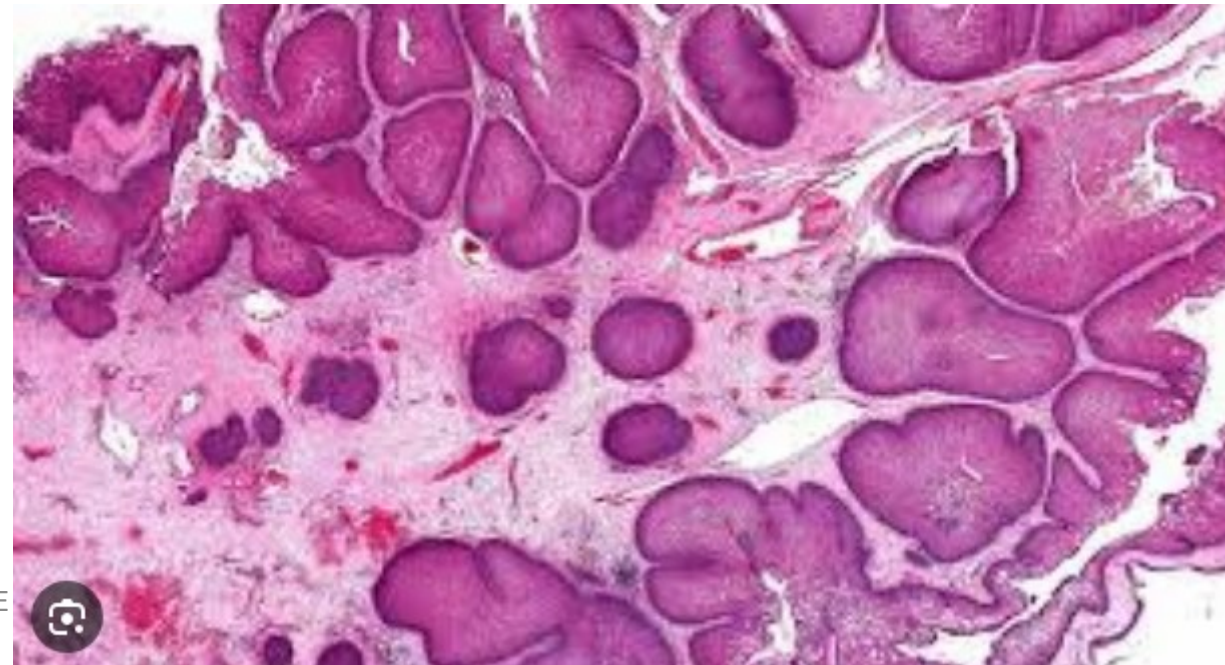
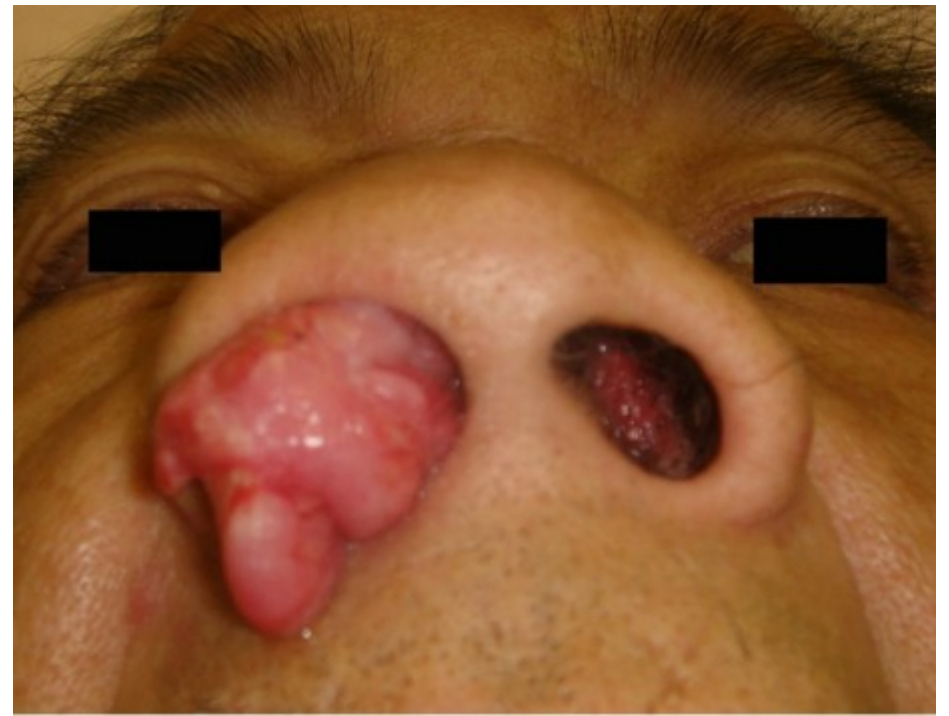
- Is a benign, highly **vascular** tumor
- **locally aggressive**, with intracranial extension
- Recurrence rates can approach **20%**, and 9% of cases can be fatal
- occurs almost exclusively in **adolescent males**
- It is believed to arise within the fibrovascular stroma of the **posterolateral wall of the roof of the nasal cavity**
- Mutations in **CTNNB1**, which encodes beta-catenin, are present in the majority
- Can develop sporadically or syndromically in association with familial adenomatous polyposis





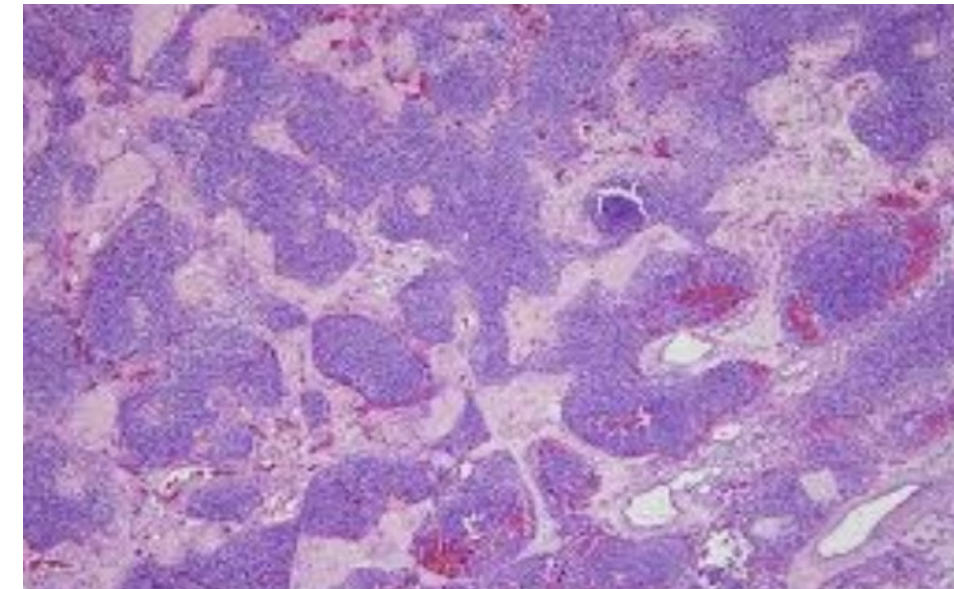
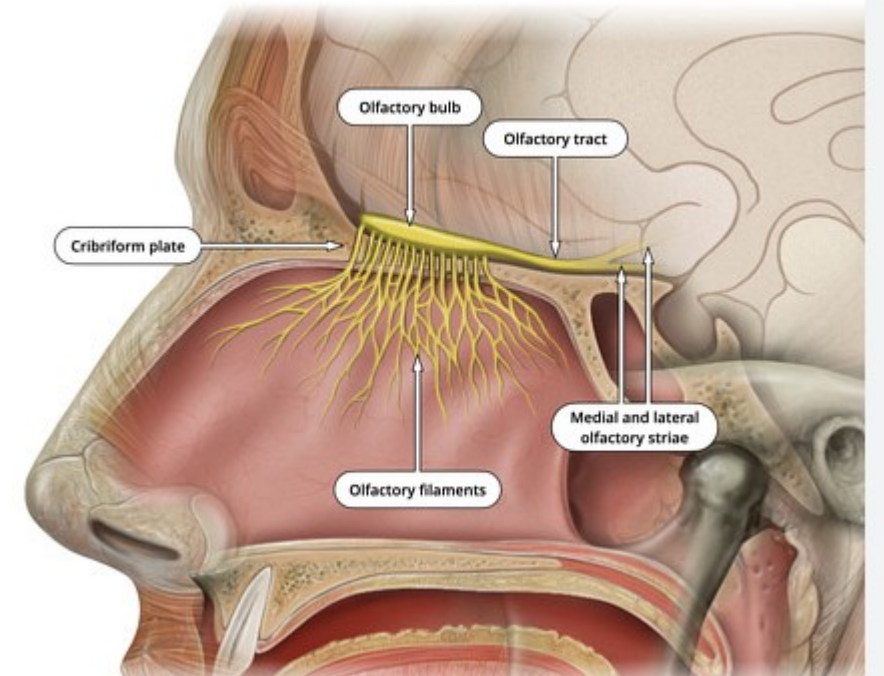
Sinonasal (Schneiderian) Papilloma

- ❖ Benign neoplasm arising from the respiratory or Schneiderian mucosa lining the nasal cavity and paranasal sinuses.
- ❖ Most common in **males** between **30 and 60** years of age.
- ❖ Occur in three forms:
 - Exophytic (most common)- Majority harbor EGFR gene mutations
 - Endophytic (or inverted)
 - Oncocytic (formerly known as cylindrical cell)
- ❖ It can display locally aggressive behavior
 - invasion into the orbit or cranial vault
 - high rate of recurrence if not adequately excised
- ❖ Malignant transformation occurs in approximately **10%** of cases



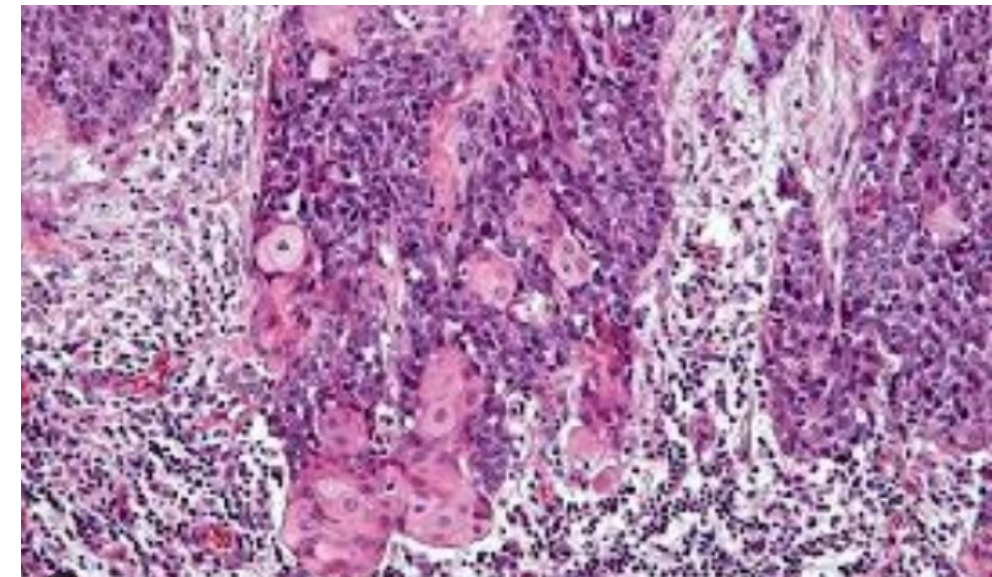
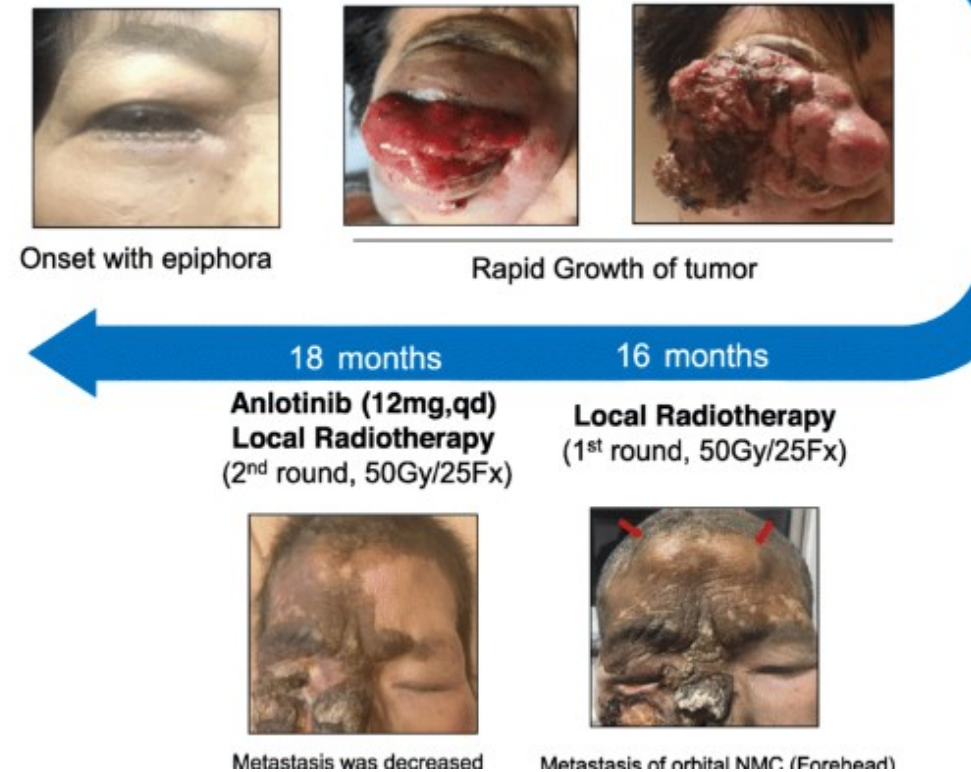
Olfactory Neuroblastoma (Esthesioneuroblastoma)

- ❖ Arise from neuroectodermal olfactory cells within the mucosa
 - superior aspect of the nasal cavity
- ❖ The age distribution is bimodal, with peaks at 15 and 50 years
- ❖ Patients typically present with
 - nasal obstruction and/or
 - Epistaxis
- ❖ Histologically, olfactory neuroblastomas are
 - **Small, blue, round cell neoplasms**
 - **Typically composed of well-circumscribed nests and lobules of cells separated by fibrovascular stroma**
 - **A fibrillary matrix, representing tangled neuronal cell processes**
- ❖ Genetic aberrations in olfactory neuroblastoma are heterogeneous



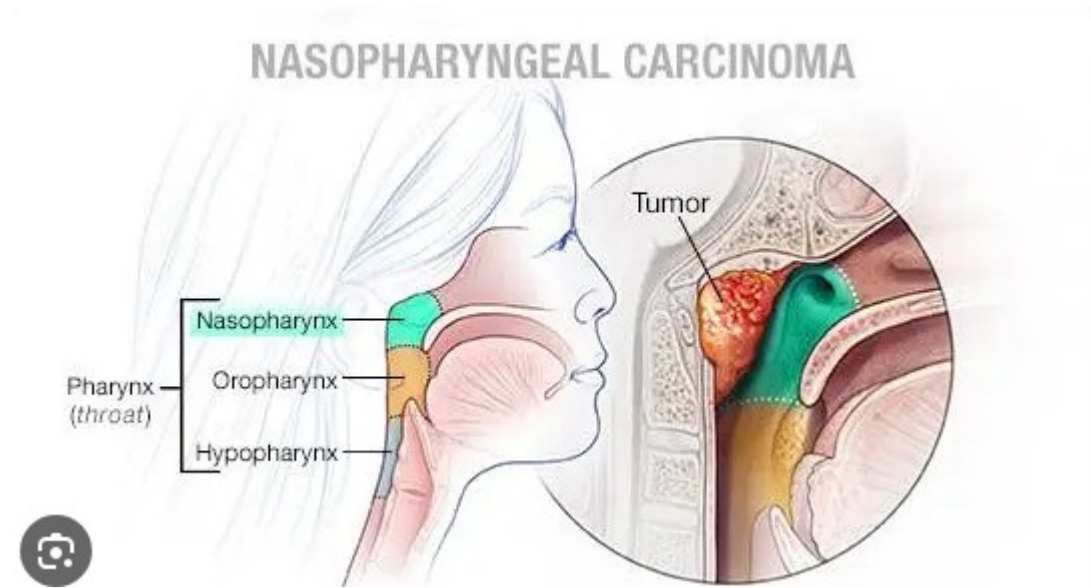
NUT Midline Carcinoma

- ❖ Uncommon tumor
- ❖ Occurs in the nasopharynx, salivary gland, or other midline structures of the thorax or abdomen
- ❖ It can occur at any age, from infancy to late adulthood
- ❖ Its true incidence is not known, as it is easily mistaken for SCC.
- ❖ Is clinically distinctive due to
 - its **extremely aggressive** behavior
 - resistance to conventional therapy
 - **most patients survive for less than 1 year following diagnosis**
- ❖ Uniformly associated with translocations that fuse genes encoding chimeric proteins composed of
 - NUT, a chromatin regulator, and
 - a portion of a “chromatin reader” protein, usually BRD4

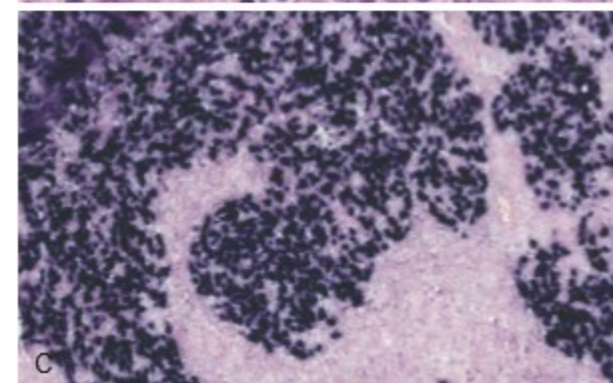
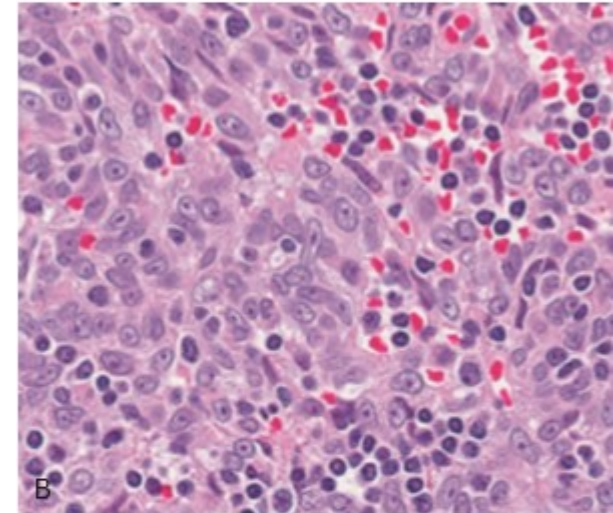
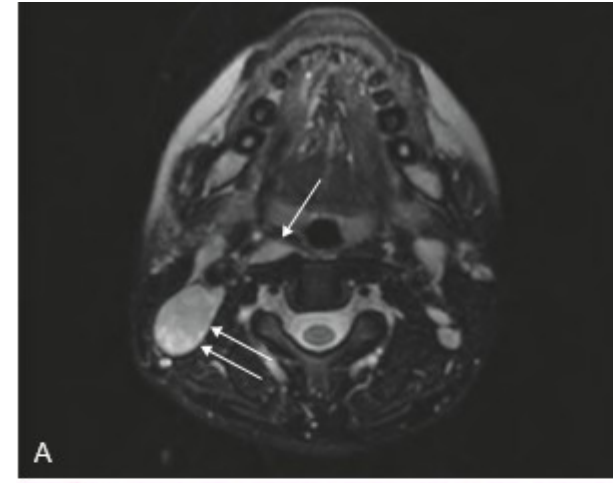


Nasopharyngeal Carcinoma

- ❖ Is characterized by a
 - distinctive geographic distribution
 - a close anatomic relationship to lymphoid tissue,
 - an association with **EBV infection**
- ❖ The nomenclature is described as one of three patterns:
 - keratinizing SCCs
 - nonkeratinizing SCCs, and
 - basaloid SCCs
- ❖ The three principal factors that influence development of nasopharyngeal carcinomas are
 - **heredity, age, and EBV infection**
- ❖ Diets high in nitrosamines, smoking and chemical fumes, have been linked to these tumor.
- ❖ In the nonkeratinizing form, most patients have antibodies against EBV
- ❖ **Primary nasopharyngeal carcinomas are often clinically occult**



- ❖ Histologically, keratinizing and nonkeratinizing NPCs resemble
 - Well-differentiated and Poorly differentiated SCCs arising in other location
 - ✓ Large epithelial cells with oval or round vesicular nuclei, prominent nucleoli, and indistinct cell borders
 - ✓ Abundant, normal-appearing lymphocytes, predominantly T cells
- ❖ **Radiotherapy** is the standard treatment
 - Nonkeratinizing NPCs have a better prognosis



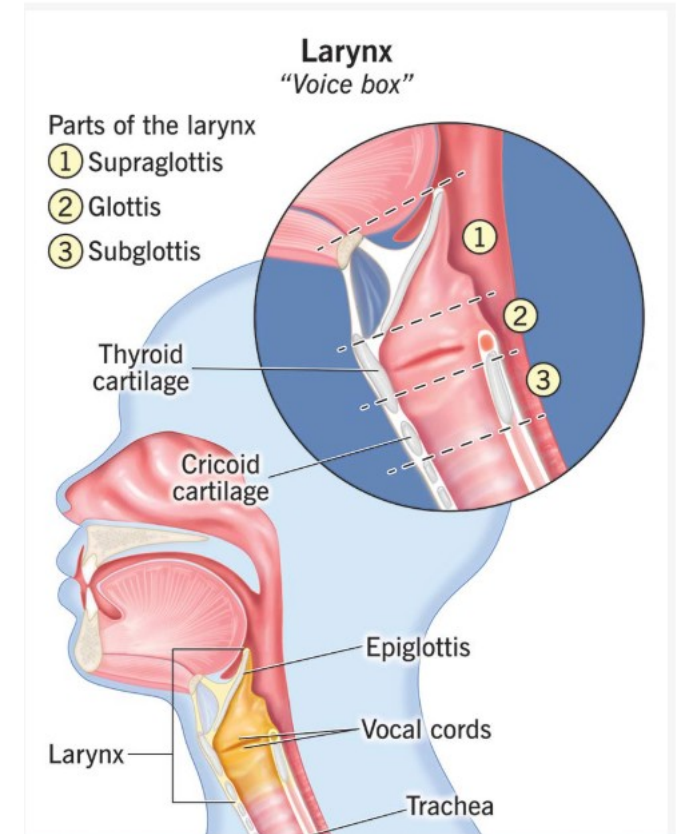
LARYNX

- ❖ The most common disorders of the larynx are inflammatory.
- ❖ Tumors are uncommon

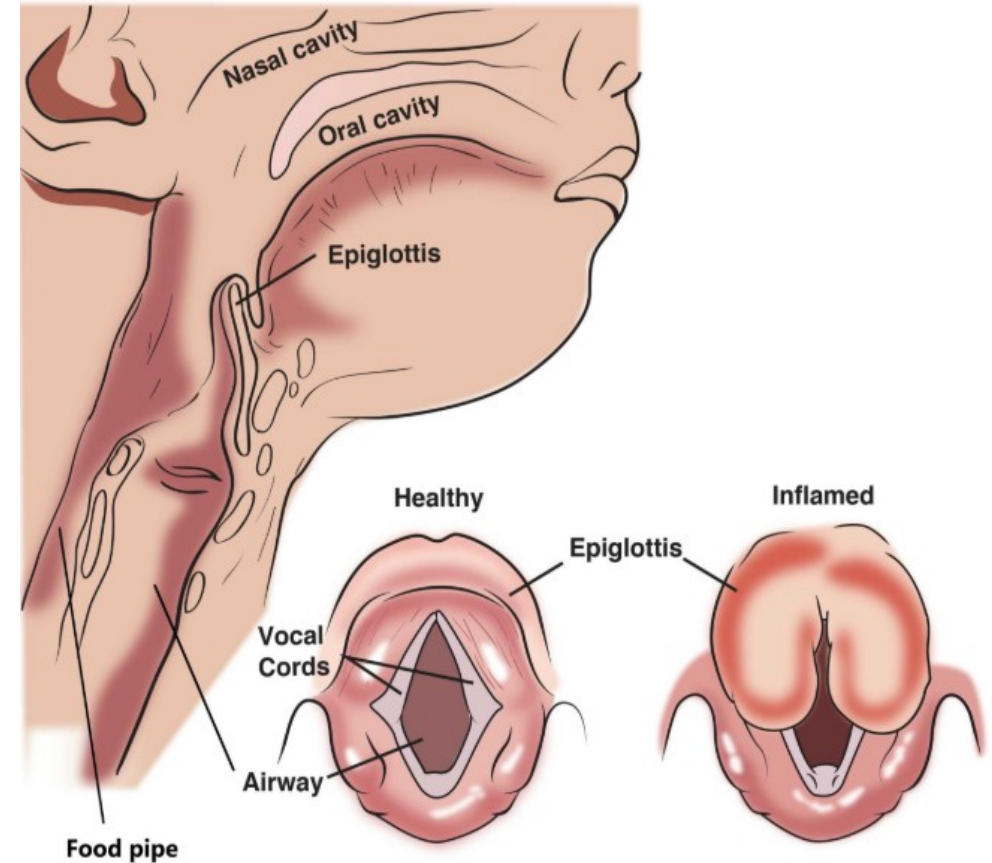
Inflammatory Lesions

Laryngitis

- ❖ May occur as the sole manifestation of allergic, viral, bacterial, or chemical insult
- ❖ More commonly associated with
 - ❖ Generalized URTI, heavy environmental toxin exposure (e.g., tobacco smoke)
 - ❖ Gastroesophageal reflux due to irritating effects of gastric contents.
- ❖ The larynx may also be affected in systemic infections, such as tuberculosis and diphtheria
- ❖ Although most infections are self-limited, the sequelae can be serious (infancy or childhood)
 - mucosal congestion, exudation, or edema may cause laryngeal obstruction.

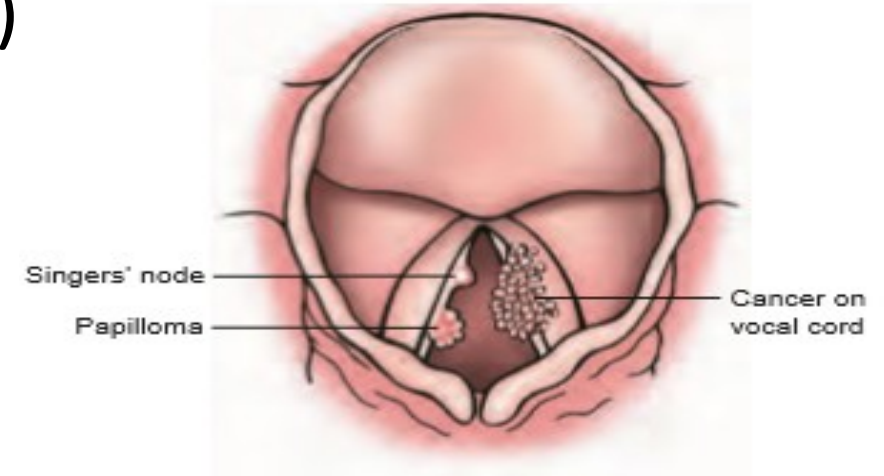


- ❖ **Laryngo-epiglottitis** caused by **respiratory syncytial virus**, **Haemophilus influenzae**, or **β-hemolytic streptococci** may induce
 - sudden swelling of the **epiglottis** and **vocal cords** as to constitute a medical emergency
- ❖ This is uncommon in adults
 - The larger size of the larynx as well as stronger accessory respiratory muscles.
- ❖ **Croup** refers to laryngo-tracheo-bronchitis in children that produces a characteristic **inspiratory stridor** due to airway narrowing.
- ❖ The laryngitis that is common in **heavy smokers** predisposes to **squamous epithelial metaplasia** and sometimes overt **carcinoma**.



Reactive Nodules (Vocal Cord Nodules and Polyps)

- ❖ Develop on the vocal cords, most often in
 - heavy smokers
 - those who impose great strain on their vocal cords (singer's nodules)
- ❖ They change the character of the voice and often cause **progressive hoarseness**
- ❖ The risk of developing cancer is almost **nonexistent**
- ❖ By convention,
 - ❖ vocal cord nodules are bilateral
 - ❖ polyps are unilateral
- ❖ Both nodules and polyps are
 - smooth, rounded, sessile or pedunculated excrescences
 - only a few millimeters in greatest dimension
 - located usually on the true vocal cords
- ❖ The histologic features
 - ❖ covered by squamous epithelium that may become keratotic, hyperplastic, or even slightly dysplastic



Squamous Papilloma and Papillomatosis

- ❖ Benign neoplasms
- ❖ Usually located on the **true vocal cords**
- ❖ Form soft, exophytic proliferations that rarely exceed 1 cm in diameter
- ❖ Usually solitary in adults but are often multiple in children(**juvenile** laryngeal papillomatosis)
- ❖ The growths are composed of
 - multiple slender, fingerlike projections supported by central fibrovascular cores
 - covered by an orderly stratified squamous epithelium
- ❖ The lesions are caused by **HPV types 6 and 11** acquired via the maternal birth canal.
- ❖ They frequently recur but only rarely transform to SCC.
- ❖ Often regress spontaneously at puberty

Carcinoma of the Larynx

- ❖ Carcinoma of the larynx is typically a **squamous cell carcinoma** seen in male chronic smokers
- ❖ Most often related to **tobacco smoke**
- ❖ Risk is proportional to exposure
- ❖ Prior to malignant transformation, epithelial changes frequently **regress** after smoking cessation
- ❖ Alcohol synergizes with tobacco to increase risk substantially

Hyperplasia-Dysplasia-Carcinoma Sequence

- ❖ Epithelial alterations in the larynx range from
 - hyperplasia to dysplasia, carcinoma in situ, and invasive carcinoma.
- ❖ Nondysplastic hyperplasias have almost no potential for malignant transformation.
- ❖ For lesions with dysplasia, the risk of malignant transformation is directly proportional to the **initial grade of dysplasia**

- ❖ About **95%** of laryngeal carcinomas are **typical SCCs**.
- ❖ Most arise on the **vocal cords**, but they may develop above or below the cords and involve the **epiglottis, aryepiglottic folds, or pyriform sinuses**.
- ❖ The initial manifestation is often persistent hoarseness, dysphagia, and dysphonia
- ❖ Prognosis is directly related to clinical stage



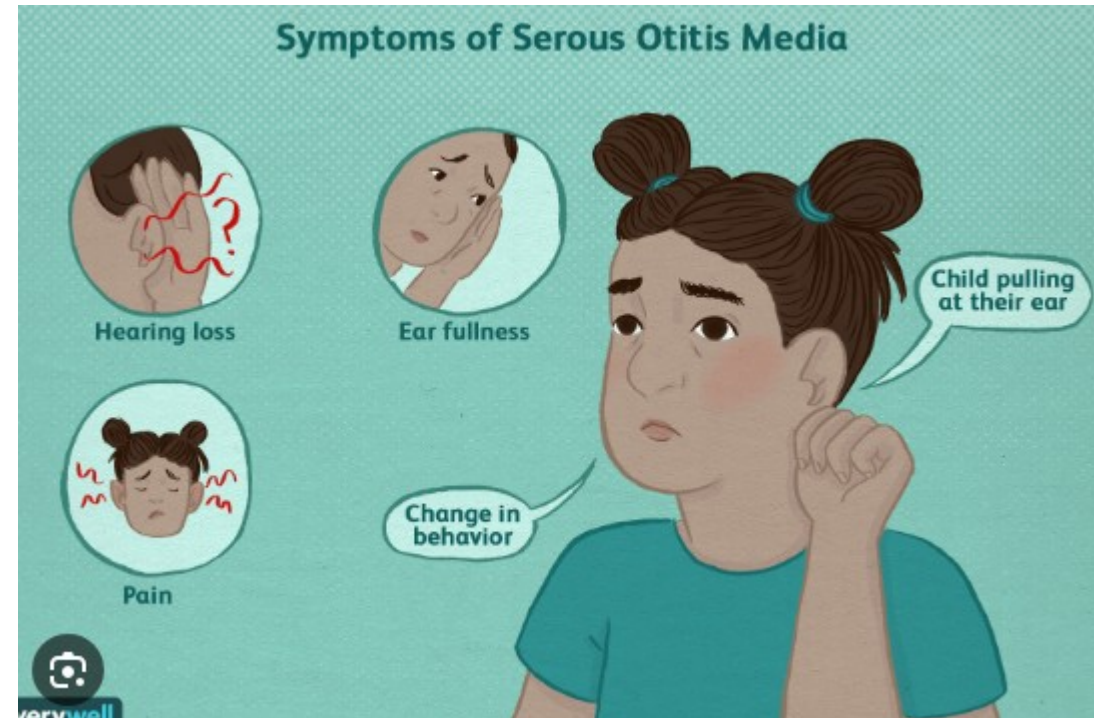
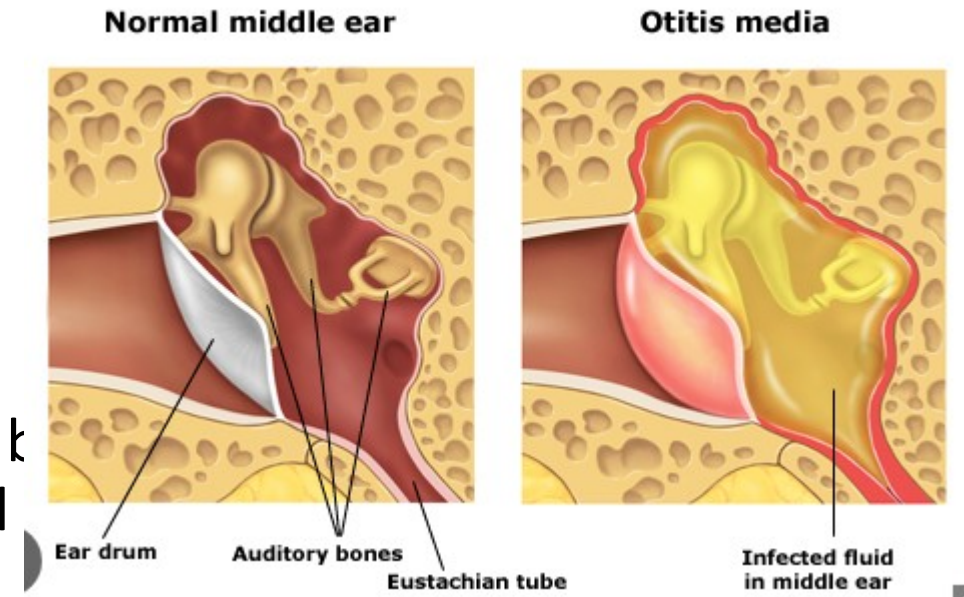
Ear

- ❖ The most common aural disorders, in descending order of frequency,
 - acute and chronic otitis (most often involving the middle ear and mastoid),
 - symptomatic otosclerosis
 - aural polyps
 - labyrinthitis
 - carcinomas, largely of the external ear; and
 - paragangliomas, found mostly in the middle ear

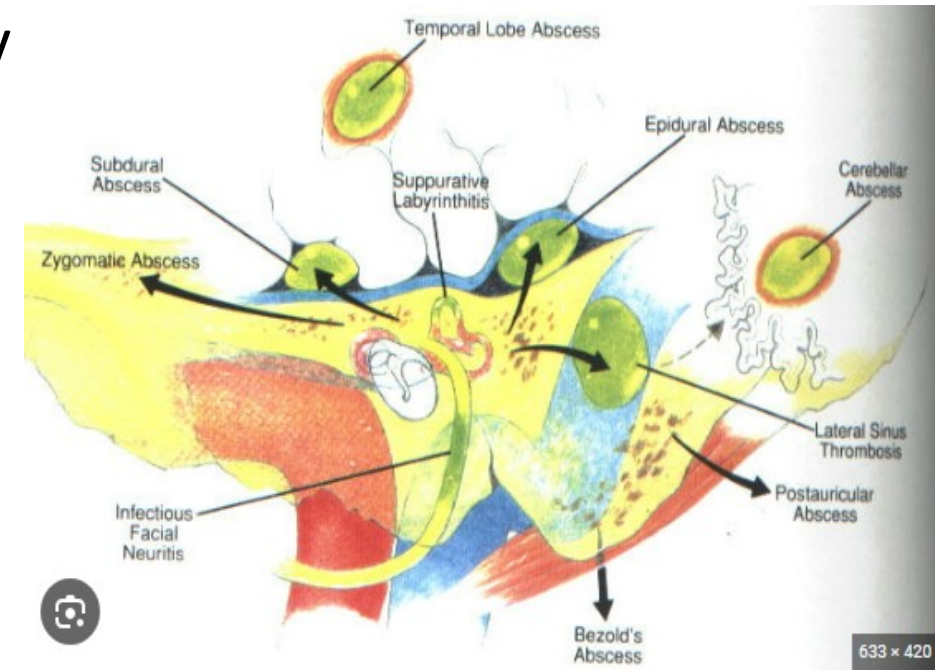
Inflammatory Lesions

Otitis media(acute or chronic)

- ❖ occur mostly in infants and children
- ❖ typically viral in nature and produce a serous exudate that may become suppurative with superimposed bacterial infection.
- ❖ The most common bacteria in the acute infection
 - Streptococcus pneumoniae
 - non-typeable H. influenzae
 - Moraxella catarrhalis

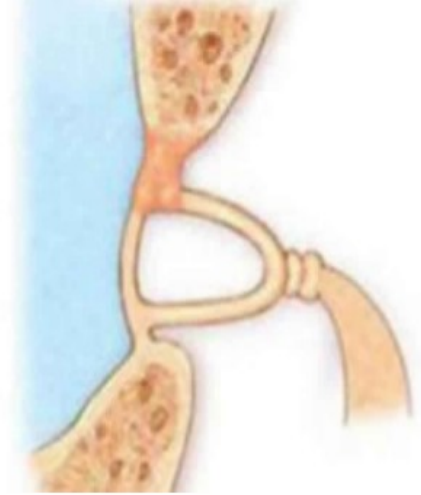
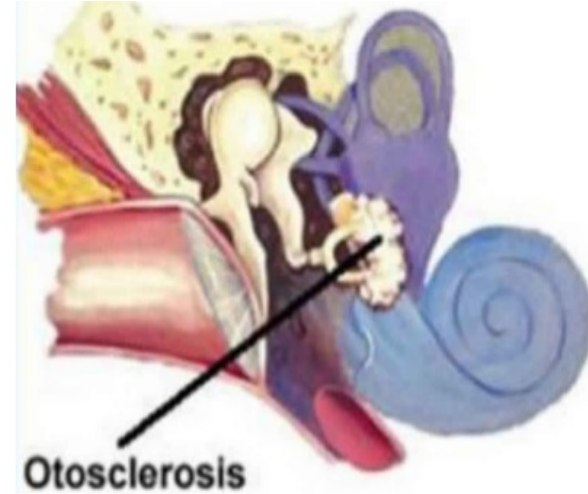


- ❖ Repeated bouts of acute otitis media with failure of resolution lead to **chronic disease**.
- ❖ The causative agents of chronic disease are usually
 - *Pseudomonas aeruginosa*
 - *Staphylococcus aureus*
 - fungus; sometimes a mixed flora
- ❖ Chronic infection has the potential to
 - perforate the eardrum
 - encroach on the ossicles or labyrinth
 - spread into the mastoid spaces
 - Penetrate into the cranial vault to produce a temporal cerebritis or abscess.
- ❖ Cholesteatomas, associated with chronic otitis media, are non-neoplastic, cystic lesions 1 to 4 cm in diameter



Otosclerosis

- Refers to abnormal bone deposition in the middle ear about the rim of the oval window
- Both ears are usually affected
- The degree of immobilization governs the severity of the hearing loss
- usually begins in the early decades of life



THANK YOU!!!